20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

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Abstract

Introduction

Switzerland's cantonal electoral reforms of the last two decades have brought renewed attention to a central question of representative democracy: how parliamentary seats can be distributed in a way that reflects voter preferences as fairly as possible while still preserving territorial representation. In the Swiss case, this debate became especially prominent because traditional proportional systems in small constituencies often generated high implicit thresholds. These electoral systems therefore dis-advantaged smaller parties which led to raising concerns about electoral equality (Bundeskanzlei, 2013, pp. 8–10). The double-proportional allocation method, commonly known as the double Pukelsheim, emerged in this context as a reform intended to reduce such distortions and to strength-en the equal weight of votes across the canton (Lüscher, 2023, p. 571).

At the core of this debate lies the contrast between two different logics of seat allocation. The tradi-tional Hagenbach-Bischoff method belongs to the family of divisor methods with downward round-ing and therefore tends to benefit larger parties, especially in smaller electoral districts. (Bun-deskanzlei, 2013, pp. 6–7). By contrast, the double Pukelsheim combines a canton-wide upper alloca-tion with a subsequent redistribution of seats to districts, thereby improving proportionality at the cantonal level and reducing the number of votes that remain without effective influence on seat allo-cation (Pukelsheim & Schuhmacher, 2011, pp. 1581–1582, 1591). In this sense, the reform is closely linked to the principle of Erfolgswertgleichheit, meaning that votes should contribute as equally as possible to the final electoral outcome (Bundeskanzlei, 2013, pp. 8–9).

This reform is especially relevant for the representation of smaller parties. Because the double Pukelsheim aggregates party strength across the entire canton before allocating seats to districts, votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level (Lüscher, 2025, pp. 1–2). Sandro Lüscher illustrates this mechanical effect with the case of Schwyz in 2016: under the old Hagenbach-Bischoff system, the Green

Liberal Party would not have won a single seat, whereas under the new method they obtained three seats (Lüscher, 2025, p. 2-3). More generally, recent work suggests that the shift away from Hagenbach-Bischoff re-duces the structural advantage previously enjoyed by larger parties and improves the parliamentary chances of smaller competitors (Bundeskanzlei, 2025, pp. 22–25). At the same time, the reform does not eliminate all tensions. While the double Pukelsheim improves party proportionality and vote equality, it may also produce new frictions at the district level. Schol-ars have pointed out that the system can weaken the direct correspondence between a district's seat share and its actual influence on the canton-wide outcome, thereby creating a tension between parti-san proportionality and territorial representation (Lüscher, 2023, pp. 577–580).

Against this background, this thesis examines the mechanical effect of electoral reform on the parlia-mentary representation of smaller parties. More specifically, it asks and shows statistcally how many seats smaller parties gain through the transition from the Hagenbach-Bischoff system to the double Pukelsheim in Swiss cantonal parliamentary elections.

The central research gap of this thesis lies in the fact that the existing literature mainly discusses the Doppelproporz in legal, normative, or system-level terms: it emphasizes improved proportionality, reduced weightless votes, and better opportunities for smaller parties, while also pointing to trade-offs such as territorial distortions and counterintuitive seat allocations. However, there is still little systematic evidence on the mechanical seat effect of the reform itself: namely, how many seats small-er parties actually gained or would have lost when comparing the observed Pukelsheim allocation with a counterfactual Hagenbach-Bischoff allocation across elections and cantons. Recent work shows that smaller parties were generally better represented under the new system and that research has increasingly focused on broader institutional effects or territorial representation, but not on a stand-ardized empirical measurement of seat gains for smaller parties across cases. The thesis fills this gap by translating the reform's abstract proportionality advantage into a concrete, comparable outcome measure: the net seat gain or loss of smaller parties under the new versus the old allocation rule.

Theory/Literature Review

Hypotheses

Research Question 20 Years of Doppelter Pukelsheim in Switzerland: What Is the Mechanical Effect of the Transition from Hagenbach-Bischoff on the Seat Allocation of Smaller Parties and Party-System Fragmentation?

Hypotheses

Hypothesis 1 H01: Across all analyzed elections, parties with less than 10% canton-wide vote share receive no more than 5% more seats in aggregate under the double-proportional Pukelsheim system than they would have received under a counterfactual Hagenbach-Bischoff allocation. $H_0:(\Sigma Seats_{(small,Pukelsheim)})/(\Sigma Seats_{(small,HB)}) \leq 1.05$

H1 (Main Hypothesis): Across all analyzed elections, parties with less than 10% canton-wide vote share receive more than 5% more seats in aggregate under the double-proportional Pukelsheim system than they would have received under Hagenbach-Bischoff. H_1: (∑Seats_(small,Pukelsheim))/(∑Seats_(small,HB))>1.05 Hypothesis 2 Option A H02a :The transition to the double Pukelsheim system did not lead to a systematic increase in party-system fragmentation. H_0,2:ENP_post-ENP_pre≤0

$H1_2$aThe transition to the double Pukelsheim system led to a systematic increase in party-system fragmentation. H_1,2:ENP_post-ENP_pre>0

ENPv=1/(∑*(i=1)*^{n}v_^{i}*2) ENPs=1/(∑*(*i=1)^{n}s_^{i}2) v = votes shares, s = seat shares

Analysis

H1: The analysis takes the actual seat allocation under the Pukelsheimer Proporz as its starting point. Based on the available election data at the municipal or district level, it then reconstructs how many seats each party would have obtained under the former Hagenbach-Bischoff system. The comparison between the actual seat distribution under the new system and the hypothetical distribution under the old system makes it possible to identify how many seats each party gained or lost as a result of the reform. In this way, the study assesses the mechanical effect of the electoral-system change on party representation, with a particular focus on smaller parties. For the purposes of this thesis, a smaller party is defined as a party that received in maximum 10% of the vote share in the relevant electoral unit. Youth lists and other special-interest lists, such as wom-en's lists or pensioners' lists, are not treated as independent parties. Instead, their votes and seats are attributed to the respective mother party, which reflects common electoral practice in Switzerland. The choice of 10% as the threshold is grounded in both legal and normative considerations. In its de-cision BGE 131 I 74, the Swiss Federal Supreme Court argued that direct and natural quorums, under-stood as barriers to representation within an electoral district, should generally not exceed 10%. The 2013 report of the Federal Chancellery refers to the same benchmark and further notes that the Ven-ice Commission considers thresholds in the range of 10–15% to be the upper acceptable limit. Using vote share rather than absolute party size is particularly appropriate because a party may be electoral-ly small in one region while being comparatively strong in another. Low thresholds are closely linked to the principle of Erfolgswertgleichheit, or equality of vote success value, which requires that votes should be translated into seats as equally and effectively as possible. Investigating the threshold is especially important, since none of the reform cantons had a <10% threshold in all districts before the implantation of the new electoral reform (Lüscher,2024, p.115).

```
# A tibble: 64 × 6
   election_id parties  vote_share small_party seats_hb seats_dp
   <chr>       <chr>         <dbl>       <int> <chr>       <int>
 1 AG_09       SVP            31.9           0 54             45
 2 AG_09       SP             15.7           0 24             22
 3 AG_09       CVP            15.0           0 23             21
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 4 AG_09       FDP         14.3              0 20             20
 5 AG_09       GRUENE       8.89             1 10             13
 6 AG_09       EVP          4.47             1 4               6
 7 AG_09       GLP          3.46             1 2               5
 8 AG_09       BDP          3.06             1 3               4
 9 AG_09       EDU          1.76             1 0               2
10 AG_09       SD           1.24             1 0               2
11 AG_09       UNS          0.0641           1 0               0
12 AG_09       L_PLUS       0.0633           1 0               0
13 AG_09       NPAK         0.0094           1 0               0
14 AG_09       FAAG         0.0086           1 0               0
15 AG_12       SVP         32.0              0 54             45
16 AG_12       FDP         15.4              0 25             22
17 AG_12       SP          15.2              0 23             22
18 AG_12       CVP         13.3              0 19             19
19 AG_12       GRUENE       7.37             1 9              10
20 AG_12       GLP          5.50             1 6               8
21 AG_12       BDP          4.36             1 1               6
22 AG_12       EVP          3.90             1 3               6
23 AG_12       EDU          1.68             1 0               2
24 AG_12       SD           0.689            1 0               0
25 AG_12       SLB          0.492            1 0               0
26 AG_12       PP           0.175            1 0               0
27 AG_12       IGG          0.0049           1 0               0
28 AG_16       SVP         31.9              0 52             45
29 AG_16       SP          18.9              0 27             27
30 AG_16       FDP         16.0              0 25             22
31 AG_16       CVP         12.1              0 17             17
32 AG_16       GRUENE       7.05             1 8              10
33 AG_16       GLP          5.26             1 7               7
34 AG_16       EVP          4.05             1 3               6
35 AG_16       BDP          2.71             1 1               4
36 AG_16       EDU          1.77             1 0               2
37 AG_16       SLB          0.172            1 0               0
38 AG_16       LOVB         0.0359           1 0               0
39 AG_20       SVP         30.3              0 52             43
40 AG_20       SP          16.5              0 25             23
41 AG_20       FDP         14.7              0 21             21
42 AG_20       CVP         12.8              0 17             18
43 AG_20       GRUENE      10.0              0 11             14
44 AG_20       GLP          9.23             1 11             13
45 AG_20       EVP          4.20             1 3               6
46 AG_20       EDU          1.60             1 0               2
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47 AG_20       LOVB         0.275            1 0              0
48 AG_20       FFF          0.238            1 0              0
49 AG_20       PPAG         0.08             1 0              0
50 AG_24       SVP         33.9              0 56            48
51 AG_24       SP          16.1              0 24            23
52 AG_24       FDP         15.3              0 23            22
53 AG_24       DIEMITTE    12.9              0 18            18
54 AG_24       GLP          8.21             1 8             11
55 AG_24       GRUENE       7.44             1 8             10
56 AG_24       EVP          3.89             1 3              5
57 AG_24       EDU          1.81             1 0              3
58 AG_24       PPAG         0.0973           1 0              0
59 AG_24       LOVB         0.0878           1 0              0
60 AG_24       FIT          0.0761           1 0              0
61 AG_24       MASSVOLL     0.0644           1 0              0
62 AG_24       DBP          0.0534           1 0              0
63 AG_24       MUSIK        0.0351           1 0              0
64 AG_24       NB           0.011            1 0              0
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# A tibble: 11 × 6
   election_id parties   vote_share small_party seats_hb seats_dp
   <chr>       <chr>          <dbl>       <int> <chr>       <int>
 1 GR_22       DIEMITTE       28.0            0 39             34
 2 GR_22       SP_GRUENE      22.7            0 21/22          27
 3 GR_22       FDP            22.0            0 30/31          27
 4 GR_22       SVP            21.1            0 24             25
 5 GR_22       GLP             6.11           1 5               7
 6 GR_26       SVP            29.4            0 39             35
 7 GR_26       DIEMITTE       23.1            0 30             28
 8 GR_26       FDP            19.4            0 27             23
 9 GR_26       SP             16.7            0 17             20
10 GR_26       GLP             5.92           1 5               7
11 GR_26       GRUENE          5.41           1 2               7
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# A tibble: 26 × 6
   election_id parties   vote_share small_party seats_hb seats_dp
   <chr>       <chr>          <dbl>       <int> <chr>       <int>
 1 NW_14       SVP           29.2             0 21             17
 2 NW_14       CVP           29.1             0 18             17
 3 NW_14       FDP           24.2             0 14             15
 4 NW_14       GRUENE        12.8             0 6               8
 5 NW_14       SP             4.78            1 1               3
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 6 NW_18       FDP           28.0             0 17             17
 7 NW_18       CVP           26.7             0 18             16
 8 NW_18       SVP           25.9             0 17             15
 9 NW_18       GRUENE        13.4             0 7               8
10 NW_18       SP             4.47            1 1               3
11 NW_18       NEMITZ         0.847           1 0               1
12 NW_18       PARTEILOS      0.605           1 0               0
13 NW_22       FDP           26.3             0 18             16
14 NW_22       DIEMITTE      25.9             0 18             15
15 NW_22       SVP           24.3             0 16             15
16 NW_22       GRUENE        11.2             0 5               7
17 NW_22       GLP            8.04            1 2               5
18 NW_22       SP             3.02            1 1               2
19 NW_22       NEMITZ         0.767           1 0               0
20 NW_22       AUFRECHT       0.549           1 0               0
21 NW_26       SVP           27.9             0 20             17
22 NW_26       DIEMITTE      25.3             0 18             15
23 NW_26       FDP           24.3             0 16             15
24 NW_26       GLP            9.03            1 4               5
25 NW_26       GRUENE         8.99            1 1               5
26 NW_26       SP             4.45            1 1               3
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# A tibble: 17 × 6
   election_id parties   vote_share small_party seats_hb seats_dp
   <chr>       <chr>          <dbl>       <int> <chr>       <int>
 1 SZ_16       SVP           32.0             0 39             33
 2 SZ_16       CVP           26.8             0 29             27
 3 SZ_16       FDP           21.4             0 22             22
 4 SZ_16       SP_GRUENE     16.7             0 10             15
 5 SZ_16       GLP            2.68            1 0               3
 6 SZ_16       EVP            0.449           1 0               0
 7 SZ_20       SVP           31.4             0 38             33
 8 SZ_20       CVP           22.9             0 26             24
 9 SZ_20       FDP           20.3             0 20             20
10 SZ_20       SP_GRUENE     18.5             0 14             17
11 SZ_20       GLP            6.90            1 2               6
12 SZ_24       SVP           38.3             0 46/47          38
13 SZ_24       DIEMITTE      23.0             0 25/26          23
14 SZ_24       FDP           18.8             0 19             19
15 SZ_24       SP_GRUENE     14.7             0 9              15
16 SZ_24       GLP            4.72            1 0               5
17 SZ_24       EVP_EDU        0.443           1 0               0
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# A tibble: 27 × 6
   election_id parties                vote_share small_party seats_hb seats_dp
   <chr>       <chr>                       <dbl>       <int> <chr>       <int>
 1 VS_17       CVP                          33.5           0 48             45
 2 VS_17       FDP                          20.2           0 29             26
 3 VS_17       SVP                          16.4           0 20             23
 4 VS_17       SP_GAUCHE_CITOYENNE          10.2           0 13             14
 5 VS_17       CSP                           8.26          1 11             10
 6 VS_17       GRUENE                        6.72          1 6               8
 7 VS_17       SP_UNABHAENGIGE               1.67          1 1               2
 8 VS_17       SP_GRUENE_UNABHAENGIGE        1.60          1 2               2
 9 VS_17       RCV                           1.45          1 0               0
10 VS_21       CVP                          31.2           0 43             40
11 VS_21       FDP                          19.7           0 28             27
12 VS_21       SVP                          16.9           0 19             22
13 VS_21       SP_GAUCHE_CITOYENNE          11.3           0 17             17
14 VS_21       GRUENE                        9.07          1 11             12
15 VS_21       CSP                           4.60          1 6               5
16 VS_21       SP_GRUENE_UNABHAENGIGE        3.30          1 3               4
17 VS_21       CSP_GLP                       2.73          1 3               3
18 VS_21       RCV                           0.616         1 0               0
19 VS_21       GLP                           0.600         1 0               0
20 VS_25       MITTE                        33.9           0 48             42
21 VS_25       SVP                          19.5           0 24             26
22 VS_25       FDP                          18.1           0 27             27
23 VS_25       SP_GAUCHE_CITOYENNE          12.0           0 18             17
24 VS_25       NEO                           6.47          1 7               7
25 VS_25       GRUENE                        6.04          1 5               8
26 VS_25       SP_GRUENE_UNABHAENGIGE        2.94          1 1               3
27 VS_25       GLP                           1.13          1 0               0
```

```
# A tibble: 22 × 6
   election_id parties  vote_share small_party seats_hb seats_dp
   <chr>       <chr>         <dbl>       <int> <chr>       <int>
 1 ZG_14       CVP          26.8             0 23             22
 2 ZG_14       SVP          23.6             0 20             19
 3 ZG_14       FDP          22.2             0 20             18
 4 ZG_14       ALG          12.8             0 8              10
 5 ZG_14       SP            9.25            1 6               7
 6 ZG_14       GLP           4.97            1 3               4
 7 ZG_14       PIRATEN       0.430           1 0               0
 8 ZG_18       CVP          25.6             0 24             21
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 9 ZG_18       SVP          22.4             0 20             18
10 ZG_18       FDP          20.9             0 18             17
11 ZG_18       ALG          13.3             0 8              11
12 ZG_18       SP           11.8             0 9               9
13 ZG_18       GLP           5.32            1 1               4
14 ZG_18       AUBUE         0.643           1 0               0
15 ZG_22       DIEMITTE     24.0             0 20             19
16 ZG_22       FDP          21.9             0 21             18
17 ZG_22       SVP          21.6             0 18             18
18 ZG_22       ALG          13.9             0 9              11
19 ZG_22       SP           10.3             0 8               8
20 ZG_22       GLP           7.47            1 4               6
21 ZG_22       EVP           0.515           1 0               0
22 ZG_22       AUFRECHT      0.392           1 0               0
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```
# A tibble: 55 × 6
   election_id parties    vote_share small_party seats_hb seats_dp
   <chr>       <chr>           <dbl>       <int> <chr>       <int>
 1 ZH_07       SVP              30.5           0 69             56
 2 ZH_07       SP               19.5           0 42             36
 3 ZH_07       FDP              16.0           0 31             29
 4 ZH_07       GRUENE           10.4           0 19             19
 5 ZH_07       CVP               7.25          1 9              13
 6 ZH_07       GLP               5.75          1 3              10
 7 ZH_07       EVP               5.23          1 6              10
 8 ZH_07       EDU               2.82          1 1               5
 9 ZH_07       SD                1.33          1 0               0
10 ZH_07       FGA_AL_PDA        1.26          1 0               2
11 ZH_07       UEBRIGE           0.0193        1 0               0
12 ZH_11       SVP              29.6           0 68             54
13 ZH_11       SP               19.3           0 45             35
14 ZH_11       FDP              12.9           0 24             23
15 ZH_11       GRUENE           10.6           0 18             19
16 ZH_11       GLP              10.3           0 17             19
17 ZH_11       CVP               4.86          1 4               9
18 ZH_11       EVP               3.78          1 0               7
19 ZH_11       BDP               3.47          1 2               6
20 ZH_11       EDU               2.57          1 1               5
21 ZH_11       FGA_AL_PDA        1.63          1 1               3
22 ZH_11       UEBRIGE           0.567         1 0               0
23 ZH_11       SD                0.405         1 0               0
24 ZH_15       SVP              30.0           0 69             54
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25 ZH_15       SP            19.7              0 44             36
26 ZH_15       FDP           17.3              0 35             31
27 ZH_15       GLP            7.64             1 11             14
28 ZH_15       GRUENE         7.22             1 10             13
29 ZH_15       CVP            4.88             1 3               9
30 ZH_15       EVP            4.27             1 3               8
31 ZH_15       FGA_AL_PDA     2.98             1 3               5
32 ZH_15       EDU            2.66             1 1               5
33 ZH_15       BDP            2.62             1 1               5
34 ZH_15       UEBRIGE        0.67             1 0               0
35 ZH_19       SVP           24.5              0 55             45
36 ZH_19       SP            19.3              0 40             35
37 ZH_19       FDP           15.7              0 29             29
38 ZH_19       GLP           12.9              0 27             23
39 ZH_19       GRUENE        11.9              0 21             22
40 ZH_19       CVP            4.29             1 2               8
41 ZH_19       EVP            4.24             1 2               8
42 ZH_19       FGA_AL_PDA     3.32             1 3               6
43 ZH_19       EDU            2.27             1 1               4
44 ZH_19       BDP            1.53             1 0               0
45 ZH_19       UEBRIGE        0.0892           1 0               0
46 ZH_23       SVP           24.9              0 57             46
47 ZH_23       SP            19.3              0 41             36
48 ZH_23       FDP           15.9              0 31             29
49 ZH_23       GLP           12.7              0 24             24
50 ZH_23       GRUENE        10.4              0 18             19
51 ZH_23       DIEMITTE       6.03             1 6              11
52 ZH_23       EVP            3.86             1 1               7
53 ZH_23       FGA_AL_PDA     2.76             1 2               5
54 ZH_23       UEBRIGE        2.19             1 0               0
55 ZH_23       EDU            1.89             1 0               3
```

Data Cleaning

To be able to analyze the data from the different elections and compare the outcomes. each dataset has to be cleaned. The data For some cantons e.g. Valais and Schwyz the data is very scattered, therefore for there multiple datasets per election have to be cleaned and merged.

The procedure for the data cleaning was the following: For each canton in alphabetically order, for each election the data was cleaned.

To ensure the compatibility of the different datasets, the column names were standardized. The column names were renamed for each dataset to the following: “total_votes”, “svp_votes”, “sp_votes”, “cvp_votes”, “fdp_votes”, “glp_votes”, “bdp_votes”, “green_votes”, “other_votes”. It

might be that some of the datasets do not contain all of these columns, since the party landscapes varies geographically and over time.

For the counterfactual, the district-level party votes and district seat magnitudes from the pre-Pukelsheim era are retained. The Hagenbach-Bischoff allocation follows the relevant cantonal law of the time (e.g., Sections 12-13 of the Aargau Grossratswahlgesetz for the 2005 election). The main counterfactual treats every party independently, ignoring list alliances, as they were abolished with the transition to Double Pukelsheim. Vote shares use each canton's *Wählerzahl* definition: district-level party votes are divided by the district seat magnitude and rounded to the nearest integer before cantonal aggregation.

The official Double Pukelsheim results (`seats_dp`) are sourced from official cantonal authorities and are documented in the Bibliography. A single lookup table consolidates all official results for easy merging with the Hagenbach-Bischoff counterfactuals. All data processing is implemented in canton-specific R scripts (R/canton_*.R) which apply shared functions for algorithm execution and validation.

Results

Interpretation of the Results

Discussion

Conclusion

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Textsources

Datasources

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Appendix

Appendix 1: Explanation Data Cleaning Process

Appendix 2: Code for Data Cleaning

Declaration of AI-Assistance